

1. Which of the following statements is/are correct for the Verilog language?

1 Point

- A. An always block sensitive to the **posedge** of a signal will be triggered if that signal transitions from 0 to x.
- B. An always block sensitive to the **negedge** of a signal will be triggered if that signal transitions from 0 to 1.
- C. An initial block executes only once and stops when it reaches the end of the block.
- D. We can use any logical operator, such as **or**, **nor**, **and**, **nand**, **xor**, etc., in the sensitivity list of an always block.

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only D
- ☐ only A, B
- ☒ only A, C
- ☐ only A, D
- ☐ only B, C
- ☐ only B, D
- ☐ only C, D
- ☐ All are correct

Yes, the answer is correct. Score: 1

Accepted answer is only A, C

2. Which of the following statements is/are correct regarding function and task in Verilog?

1 Point

- A. A function can model both combinational and sequential circuits.
- B. A function can model delays using **posedge**, but not using **negedge**.
- C. A function must have 1 output, while a task can have 0, 1, or many outputs.
- D. A task can call a function, while a function cannot call a task.

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only D
- ☐ only A, B
- ☐ only A, C
- ☐ only A, D
- ☐ only B, C
- ☐ only B, D
- ☒ only C, D
- ☐ All are correct

Yes, the answer is correct. Score: 1

Accepted answer is only C, D

3. Which of the following statements is/are correct regarding the assign statement in Verilog?

1 Point

- A. An assign statement models combinational circuit elements and not sequential circuit elements.
- B. An assign statement provides values to nets in Verilog.
- C. An assign statement can be written within an always block.

- ☐ only A
- ☐ only B
- ☐ only C
- ☒ only A, B
- ☐ only A, C
- ☐ only B, C
- ☐ All statements are correct
- ☐ No statement is correct

Yes, the answer is correct. Score: 1

Accepted answer is only A, B

4. Which of the following statements is/are correct regarding blocking and non-blocking assignments in Verilog?

1 Point

- A. The symbol of blocking assignment is = and of non-blocking assignment is <=
- B. The LHS and RHS of a blocking assignment are executed at two separate locations in the stratified event queue.
- C. For a non-blocking assignment, the execution of the next statement is blocked until the current statement is fully executed.
- D. We can use a blocking assignment inside an always block, but we cannot use a non-blocking assignment inside an always block.

- ☒ only A
- ☐ only B
- ☐ only C
- ☐ only D
- ☐ only A, B
- ☐ only A, C
- ☐ only A, D
- ☐ only B, C
- ☐ only B, D
- ☐ only C, D
- ☐ All are correct

Yes, the answer is correct. Score: 1

Accepted answer is only A

5. What is the internal representation of the integer 8'bz111 in Verilog?

1 Point

- ☐ 0000 z111
- ☐ 0000 0111
- ☐ xxxx z111
- ☒ zzzz z111

Yes, the answer is correct. Score: 1

Accepted answer is zzzz z111

6. Assume that $A = \{1011\}$. If we write, $Y = |A|$, what will be the value of Y?

1 Point

- ☐ 0
- ☒ 1
- ☐ x
- ☐ z

Yes, the answer is correct. Score: 1

Accepted answer is 1

7. Which of the following statements is/are correct for Verilog?

1 Point

- A. The statements within a **begin-end** block are executed sequentially.
- B. The statements within a **fork-join** block are executed concurrently.
- C. We cannot write a non-blocking assignment within a **begin-end** block because they must be executed concurrently.

- ☐ only A
- ☐ only B
- ☐ only C
- ☒ only A, B
- ☐ only A, C
- ☐ only B, C
- ☐ All statements are correct
- ☐ No statement is correct

Yes, the answer is correct. Score: 1

Accepted answer is only A, B

8. Which of the following statements is/are correct for coverage metrics for an RTL design?

1 Point

- A. Code coverage identifies sections of given RTL code that did not execute during simulation.
- B. Functional coverage identifies whether the specified features in the given coverage model execute during simulation.
- C. Extracting functional coverage metrics requires more effort, as we need to develop a separate coverage model to specify the design features to be verified.
- D. Code coverage is a good metric for monitoring the progress of the verification effort.

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only D
- ☐ only A, B
- ☐ only A, C
- ☐ only A, D
- ☐ only B, C
- ☐ only B, D
- ☐ only C, D
- ☒ All are correct

Yes, the answer is correct. Score: 1

Accepted answer is All are correct

9. Which of the following statements is/are correct for RTL simulators?

1 Point

- A.Event-driven simulation involves computing a chain of events and their responses until no further events are available.
- B.Cycle-based simulators evaluate the sequence of next states at successive clock edges as a function of the present state and the present input.
- C.Cycle-based simulators are typically employed to simulate asynchronous circuits because they don't have a periodic clock signal.

- ☐ only A
- ☐ only B
- ☐ only C
- ☒ only A, B
- ☐ only A, C
- ☐ only B, C
- ☐ All statements are correct.
- ☐ No statement is correct.

Yes, the answer is correct. Score: 1

Accepted answer is only A, B

10. Which of the following statements is/are correct regarding races in Verilog simulation?

1 Point

- A.Race is employed to increase the speed of simulation.
- B.We should introduce race conditions in Verilog simulation to better match simulation and synthesis results.
- C.If we don't have a race condition in RTL simulation, it implies that the coverage is less than 50%.

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only A, B
- ☐ only A, C
- ☐ only B, C
- ☐ All statements are correct.
- ☒ No statement is correct.

Yes, the answer is correct. Score: 1

Accepted answer is No statement is correct.